04: Multi-Master Buses

Engr 315: Hardware / Software Codesign

Andrew Lukefahr
Indiana University

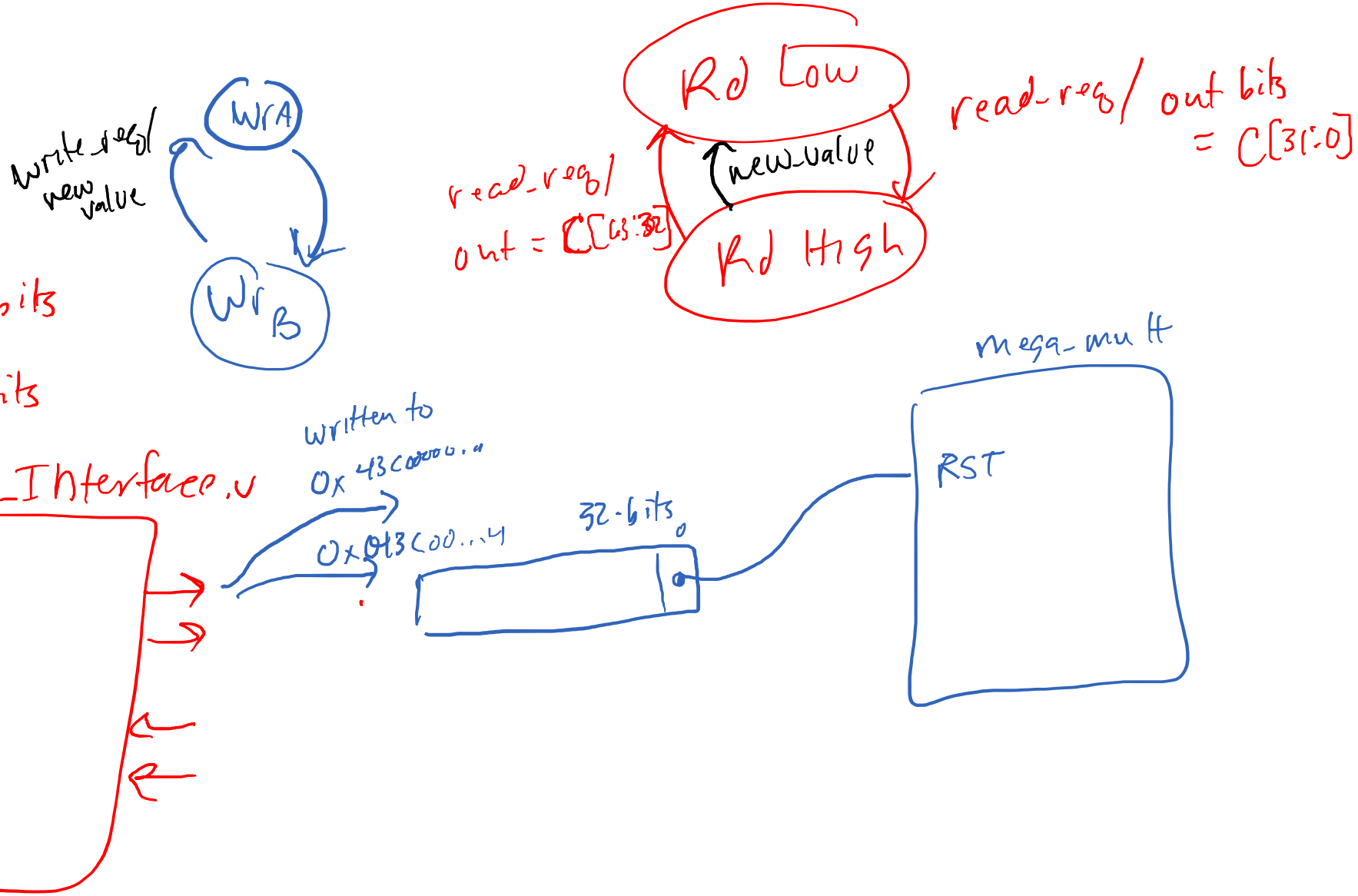
Let’s talk P2

32 bit * 32-bit = 64 bit

1 register*

Let's talk P2

- 0x43C00000 - 32-bits
- 0x43C00004 - 32 bits



Other Announcements

Slack:
email
→ zwlevy@iu.edu

- Ethan's Office Hours schedule

Wednesday	5.30 – 8:00 pm
Thursday	2.30 – 4:00 pm

- New FPGA boxes are ordered

ARM AXI Bus

- "Advanced eXtensible Interface" Bus Version 4, "AXI4"

- Three Variants
  - AXI4: Fast but complicated; Memory-mapped
  - AXI4 Lite: Slow but simple; Memory-mapped
  - AXI4 Stream: Fast and simple; Not memory-mapped

High-Performance Bus Ideas

- Make single transaction faster
- Overlap multiple transactions

AXI Handshake Speedup

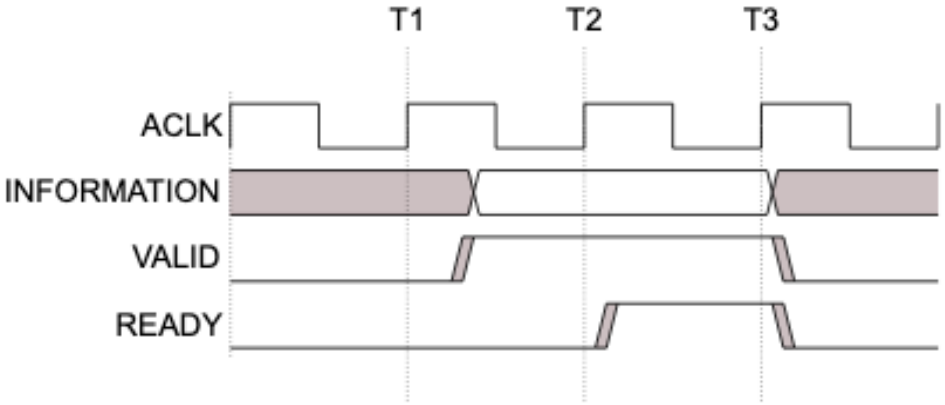


Figure A3-2 VALID before READY handshake

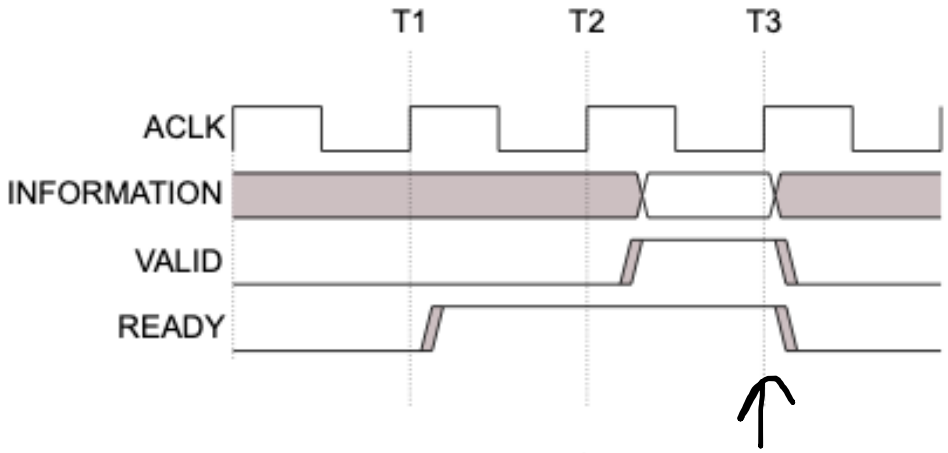


Figure A3-3 READY before VALID handshake

- Both are valid
- Figure A3-3 is faster

Burst Transaction

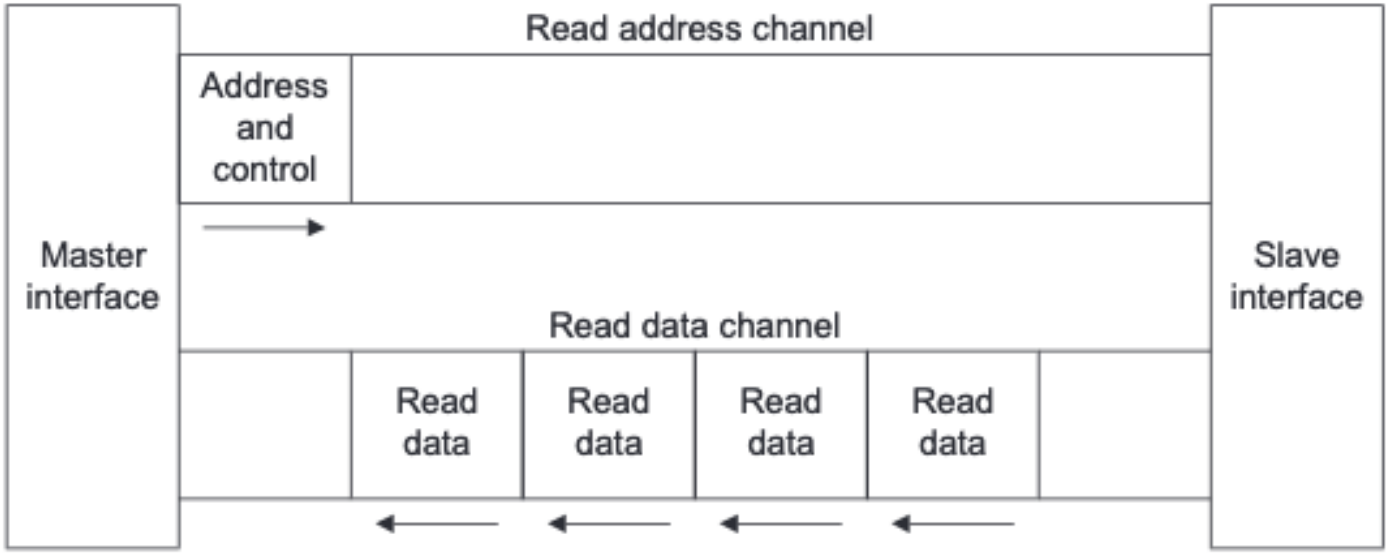
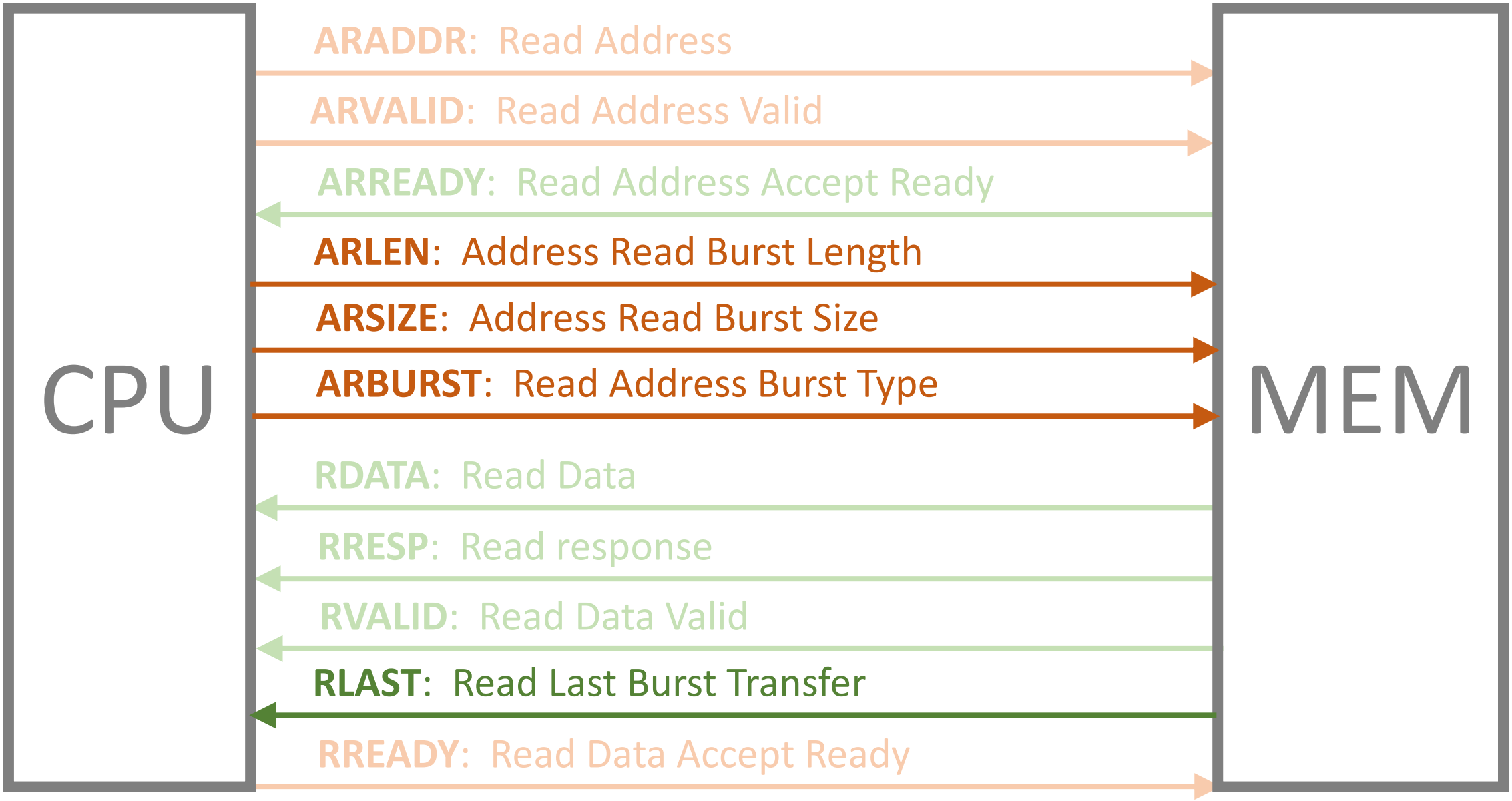
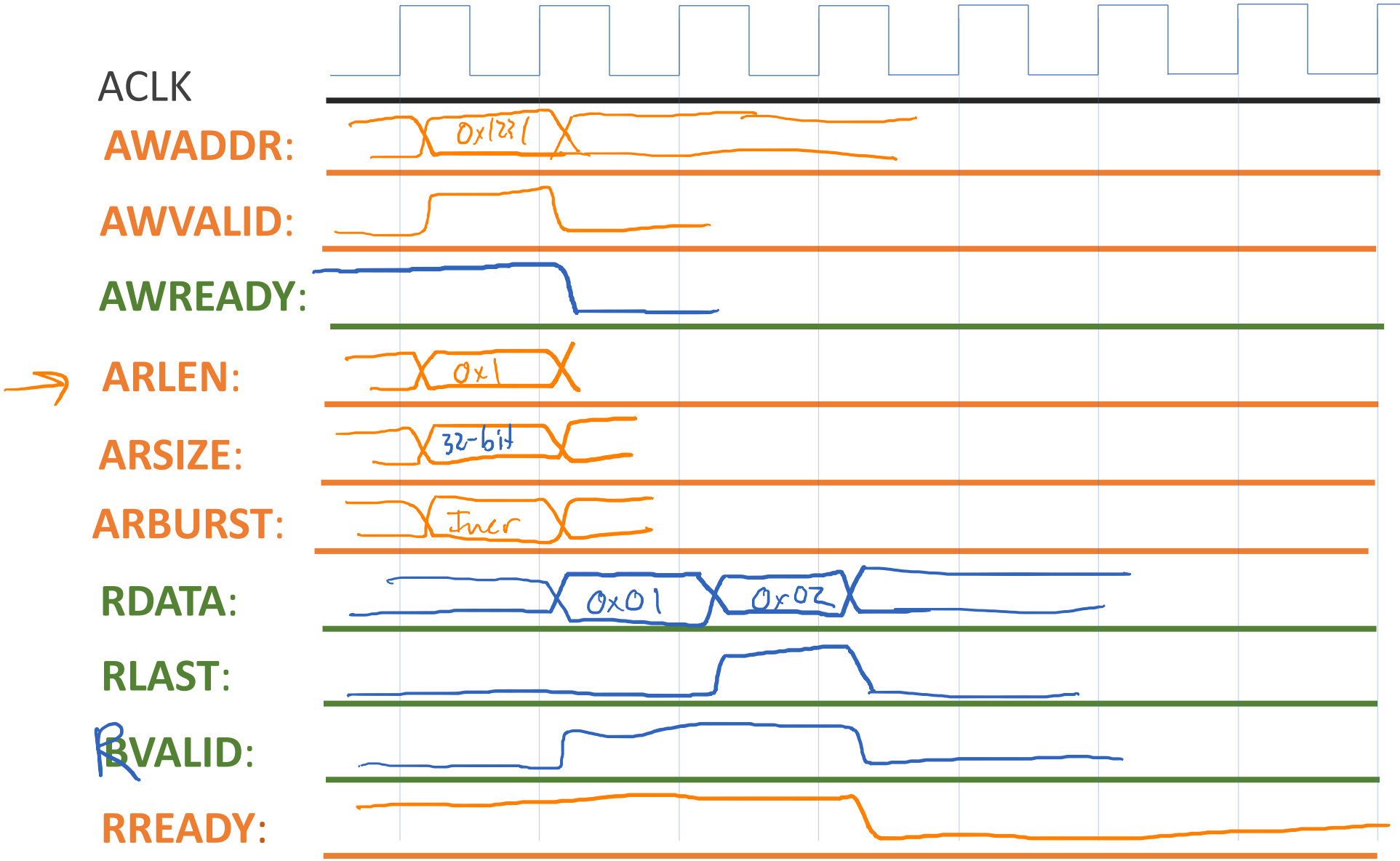


Figure 1-1 Channel architecture of reads


CPU
MEM
ARADDR: Read Address
ARVALID: Read Address Valid
ARREADY: Read Address Accept Ready
ARLEN: Address Read Burst Length
ARSIZE: Address Read Burst Size
ARBURST: Read Address Burst Type
RDATA: Read Data
RRESP: Read response
RVALID: Read Data Valid
RLAST: Read Last Burst Transfer
RREADY: Read Data Accept Ready

Reading 0x1234 and 0x1238



ARESETN and RRESP omitted

Overlapping Read Burst Transactions

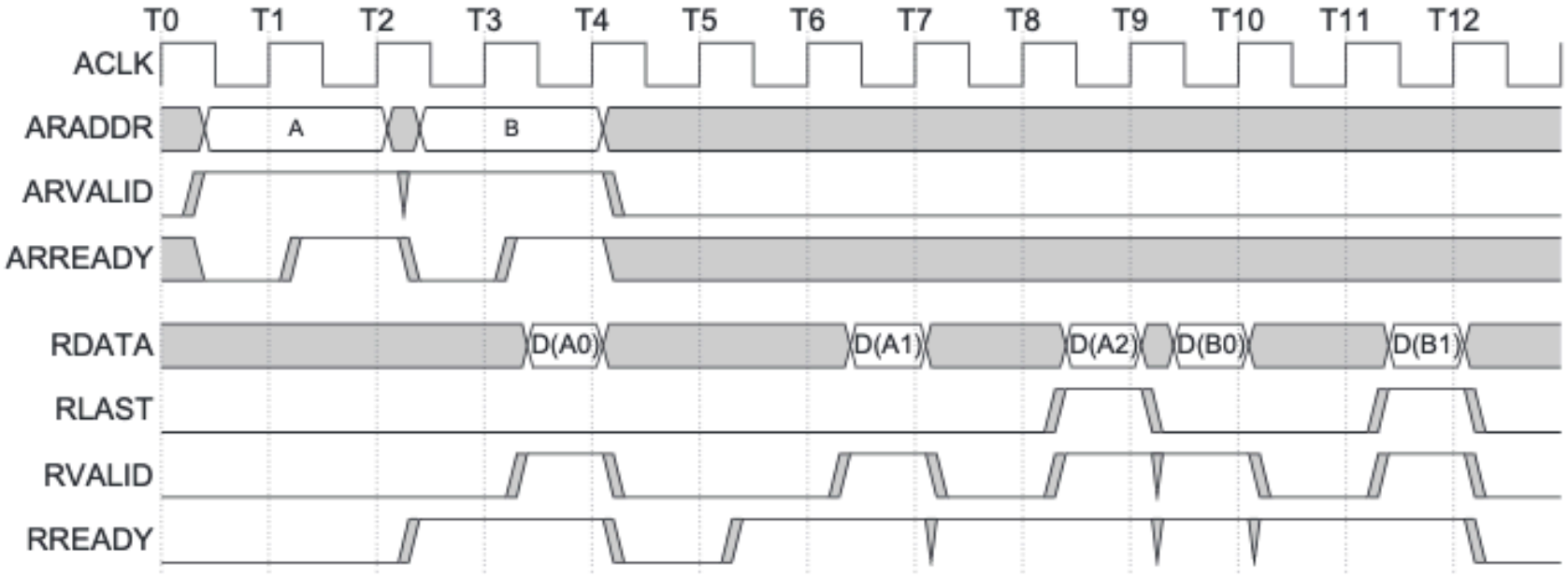


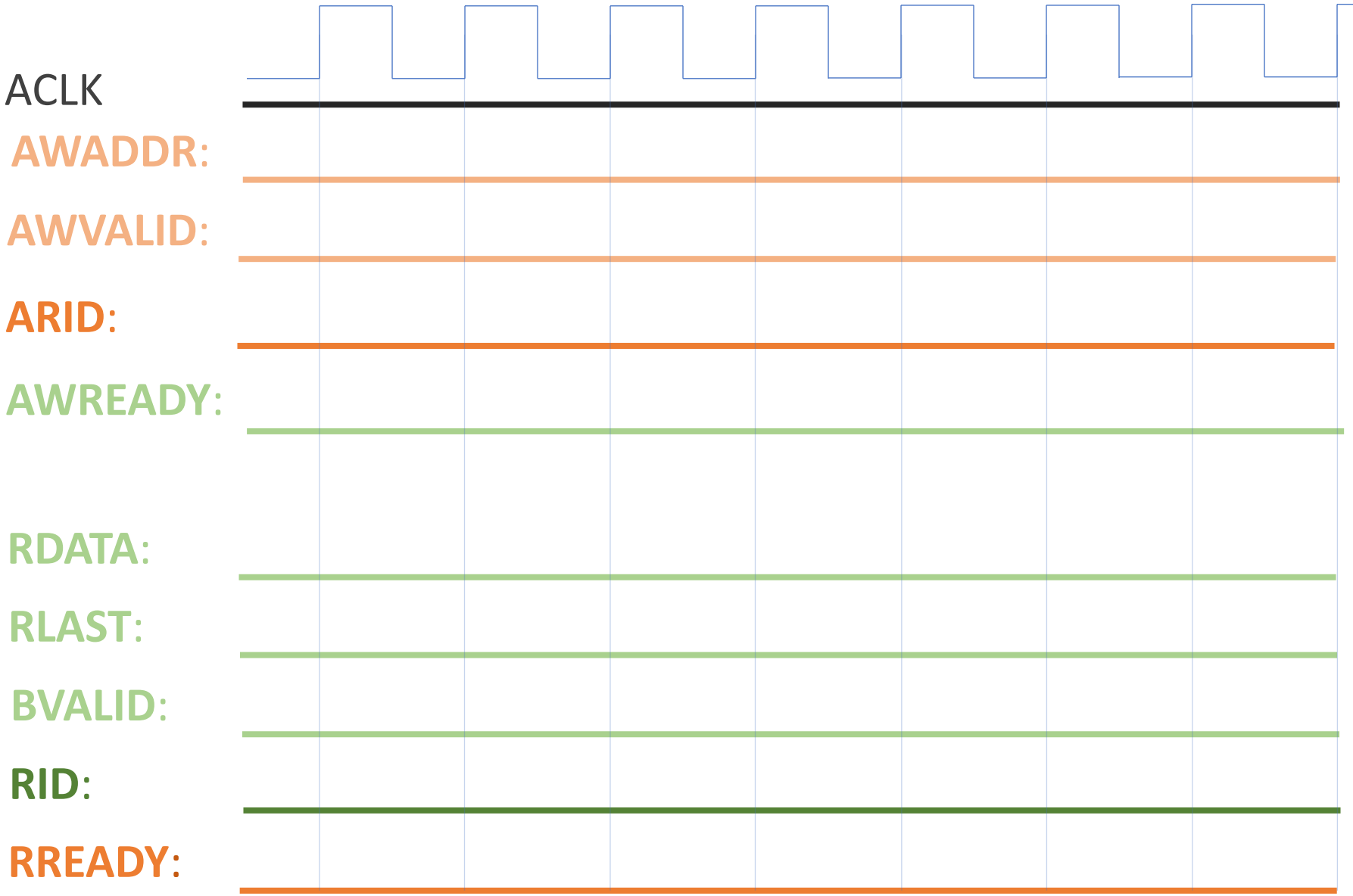
Figure 1-5 Overlapping read bursts

Re-ordering Read Transactions

- Give each transaction an “Identifier” or ID
- More new signals: AWID/WID/BID and ARID/RID

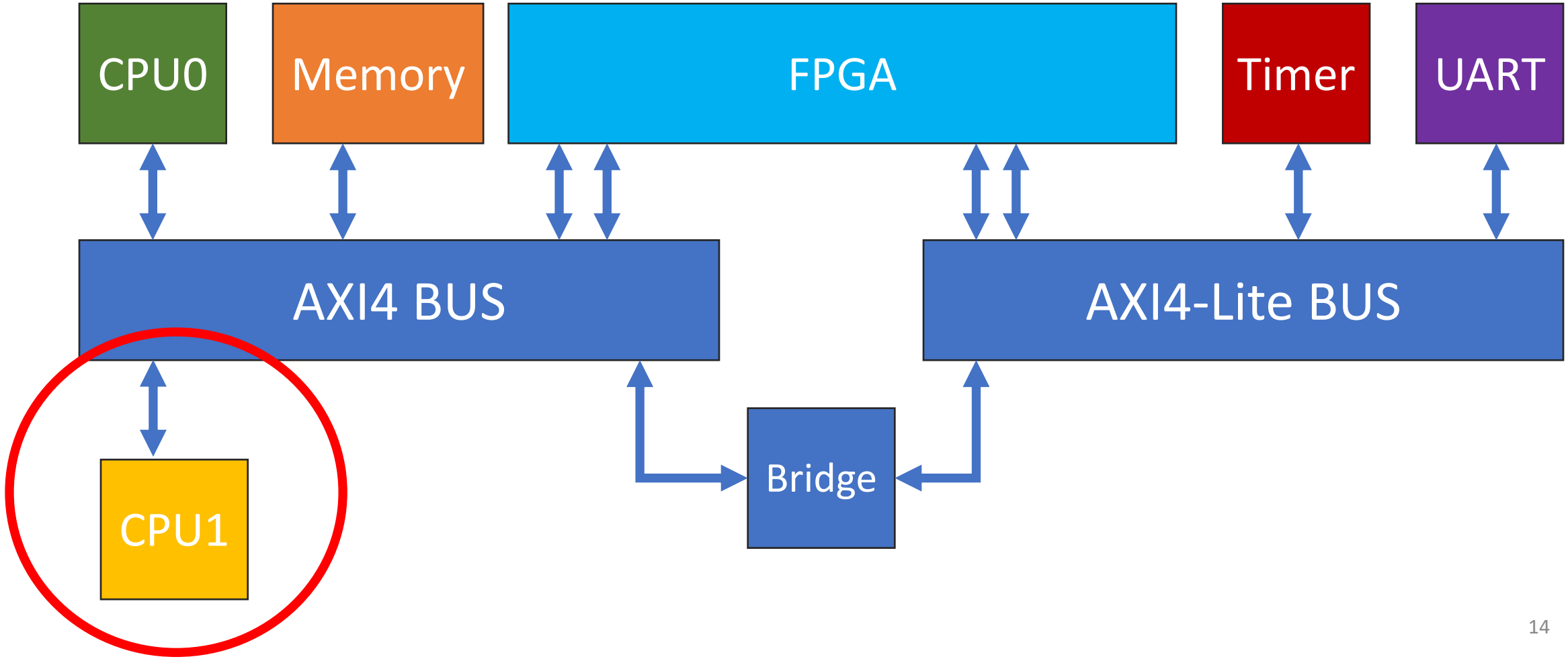
- “Transactions from the same master, but with different ID values, have no ordering restrictions. They can complete in any order.”

Reading 0x1000, 0x2000, & 0x3000

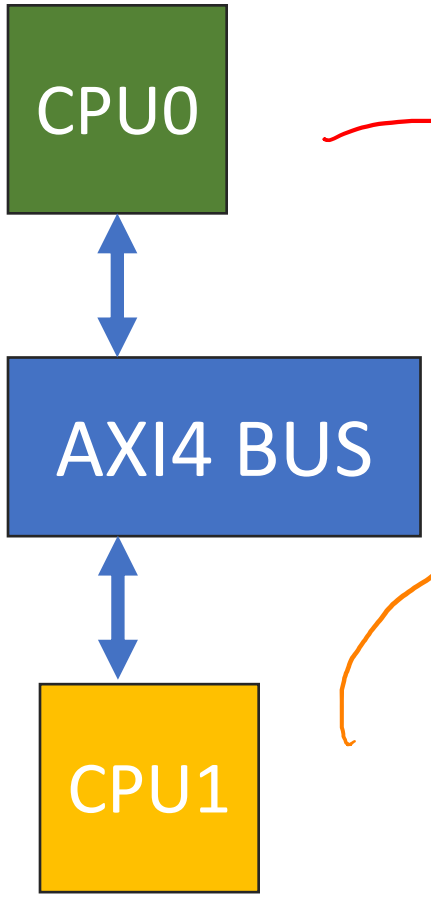


ARESETN,
ARLEN, ARSIZE, &
ARBURST omitted

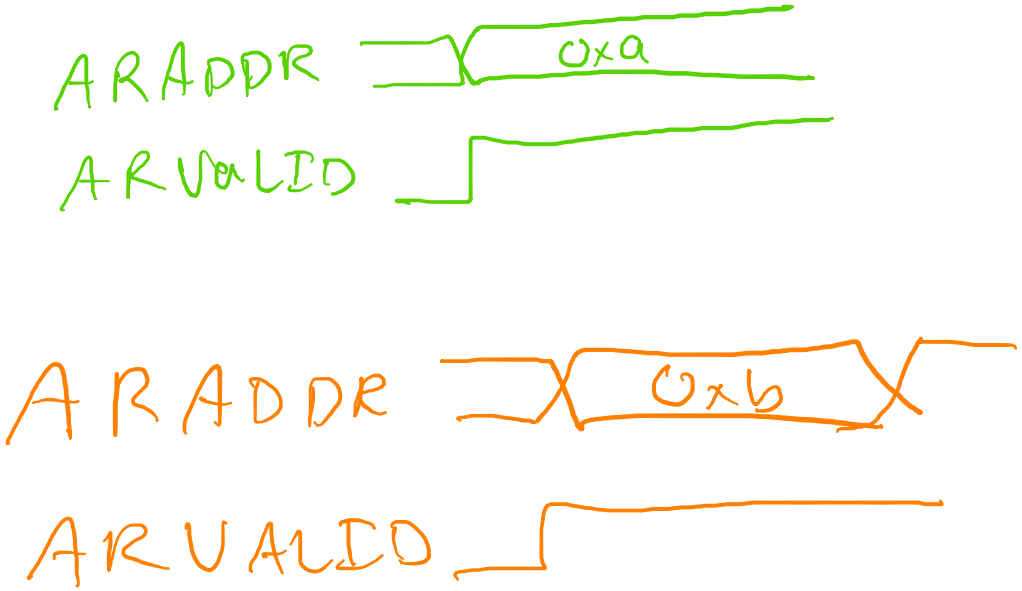
Multiple Masters



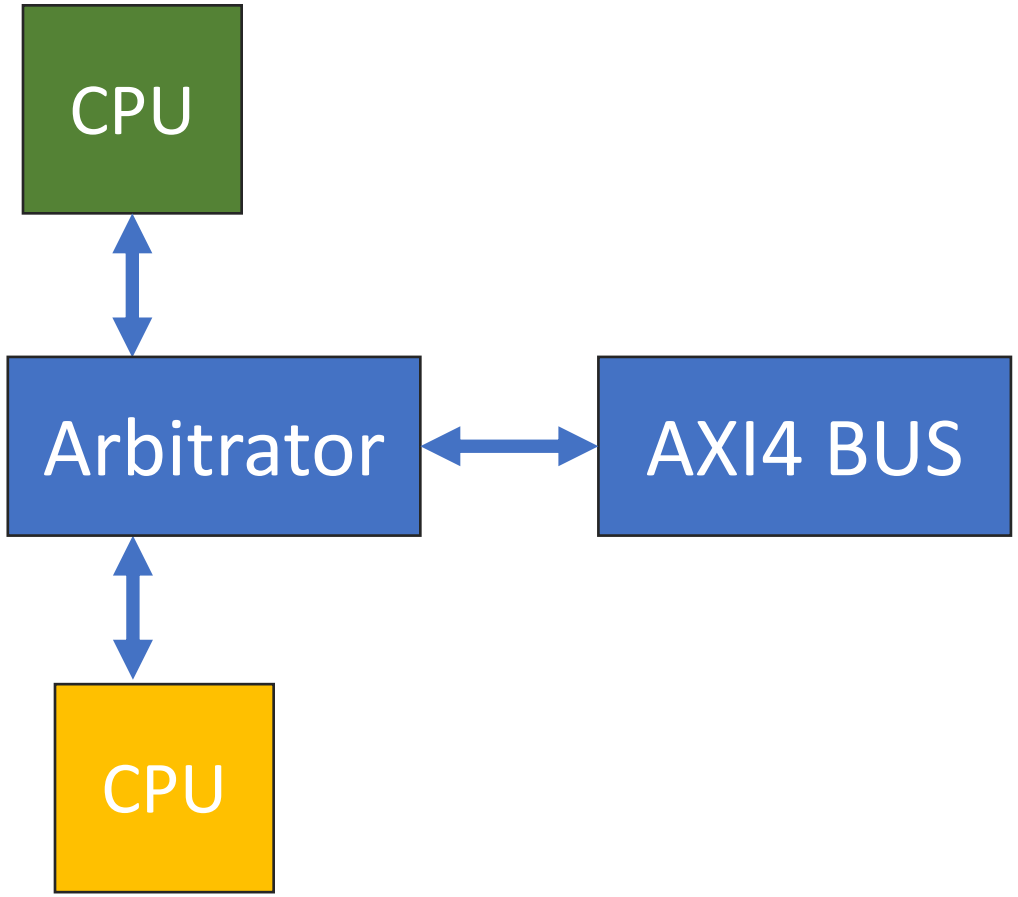
Multi-Master Buses



- What happens if both request a transaction at the same time?



An Arbitrator selects who gets to use the bus



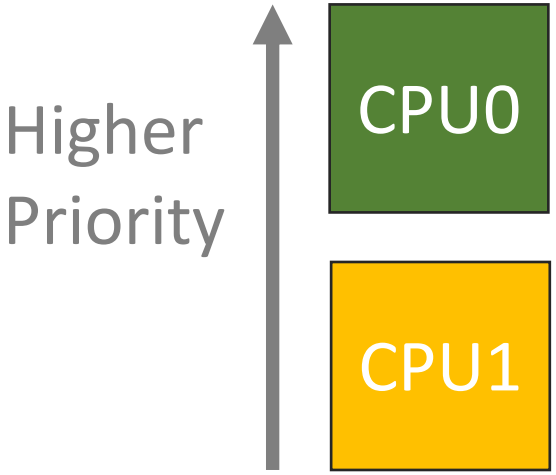
- What happens if both request a transaction at the same time?

- Arbitration: We pick a winner!

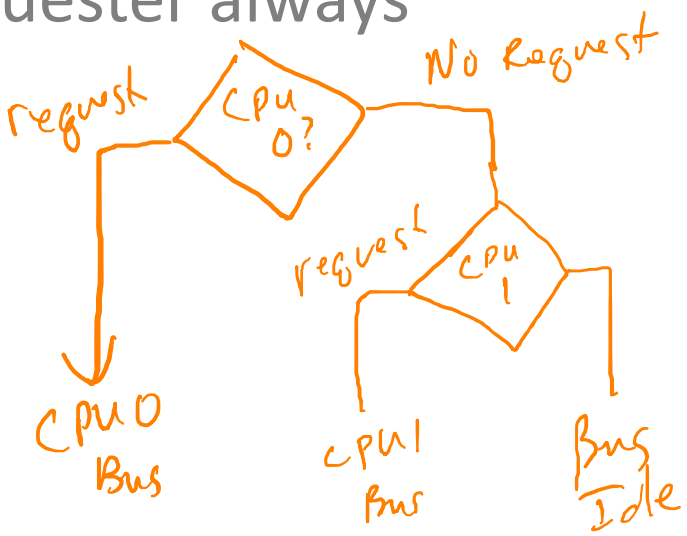
Arbitration Options

- Highest-Priority First
- Round Robin

Highest-Priority First

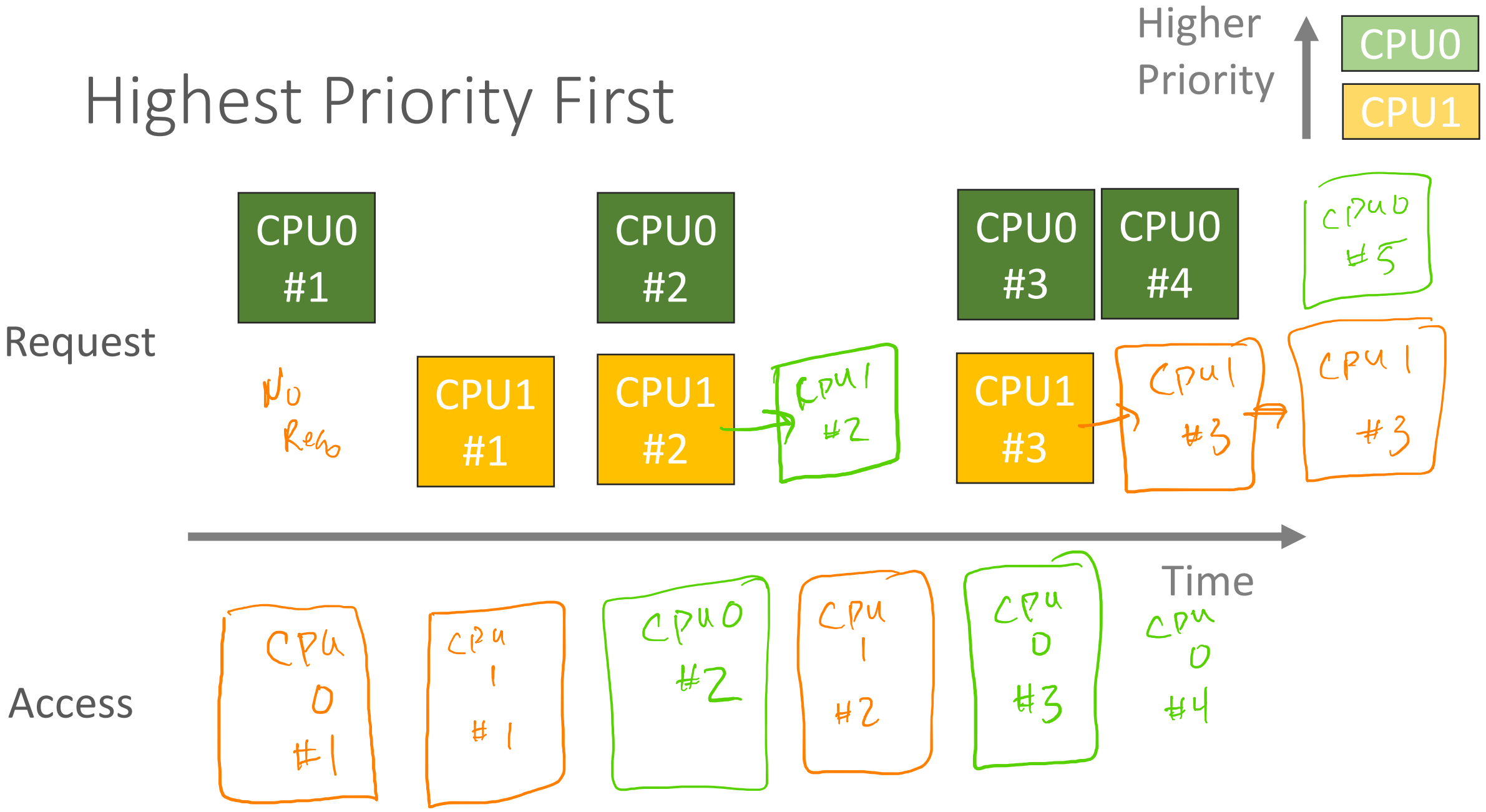


- Priority Fixed. Highest priority requester always wins.

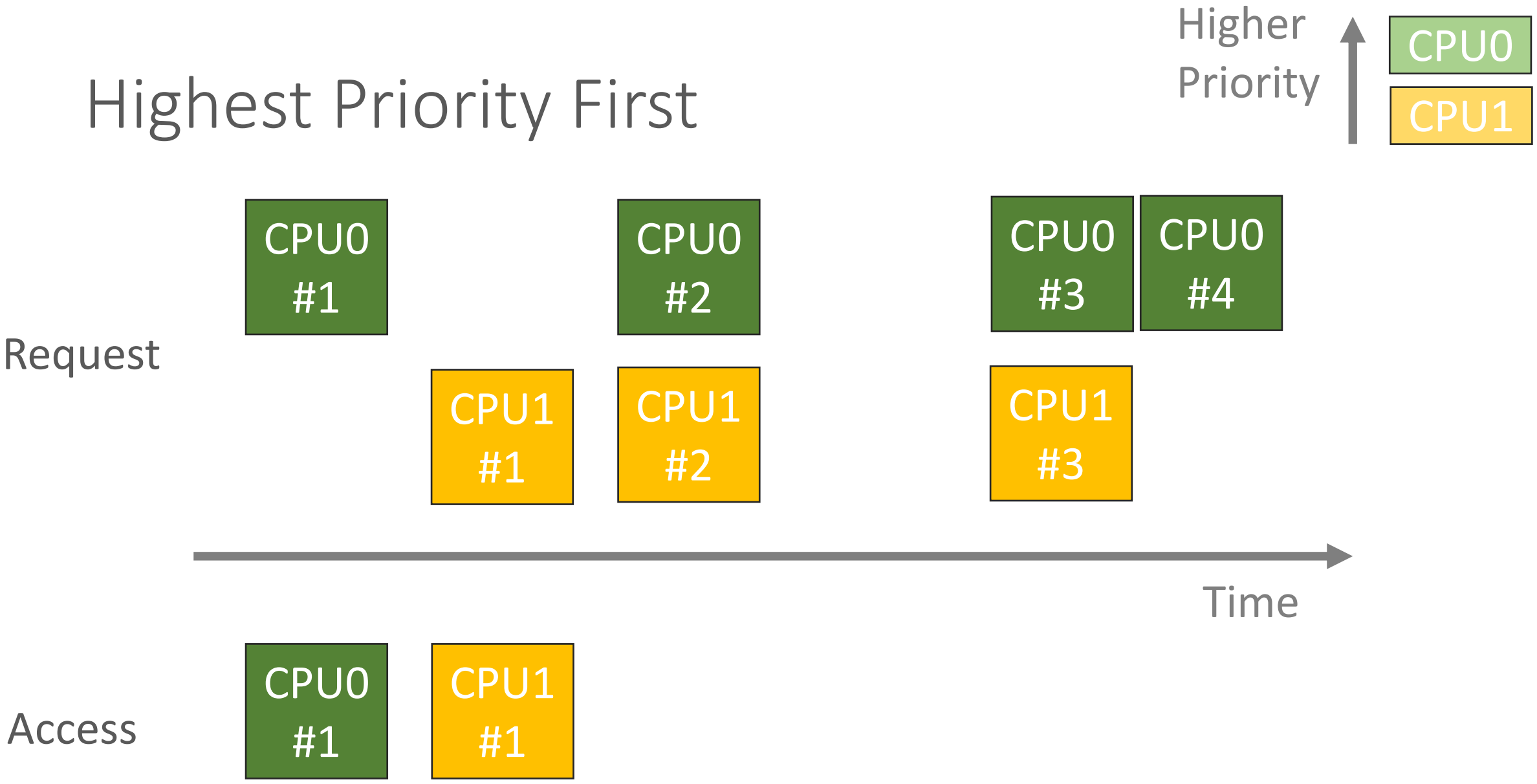


- Rules:
  - If CPU0 requests, CPU0 gets bus
  - Else if CPU0 does not request and CPU1 requests, CPU1 gets bus
  - Else bus idle

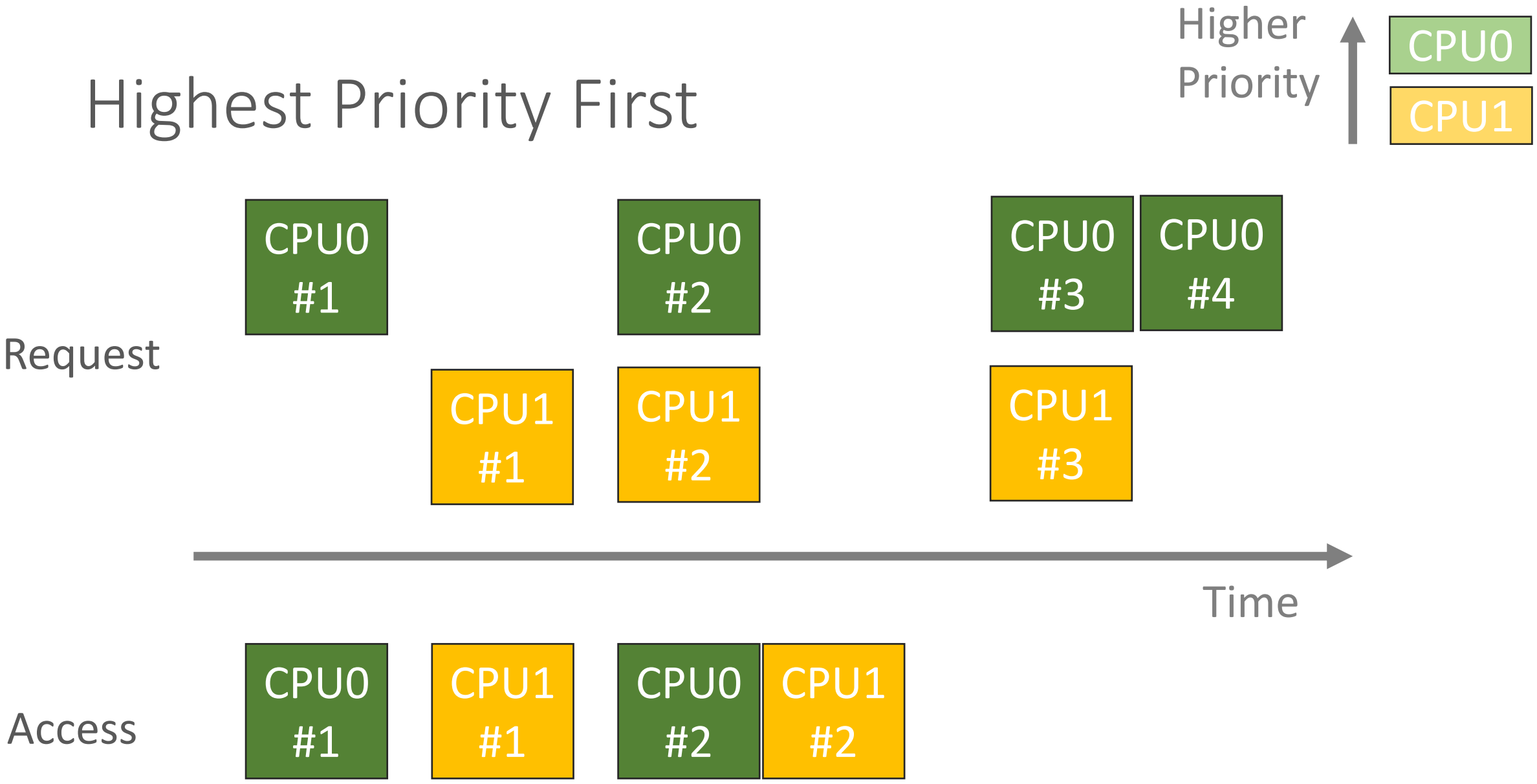
Highest Priority First



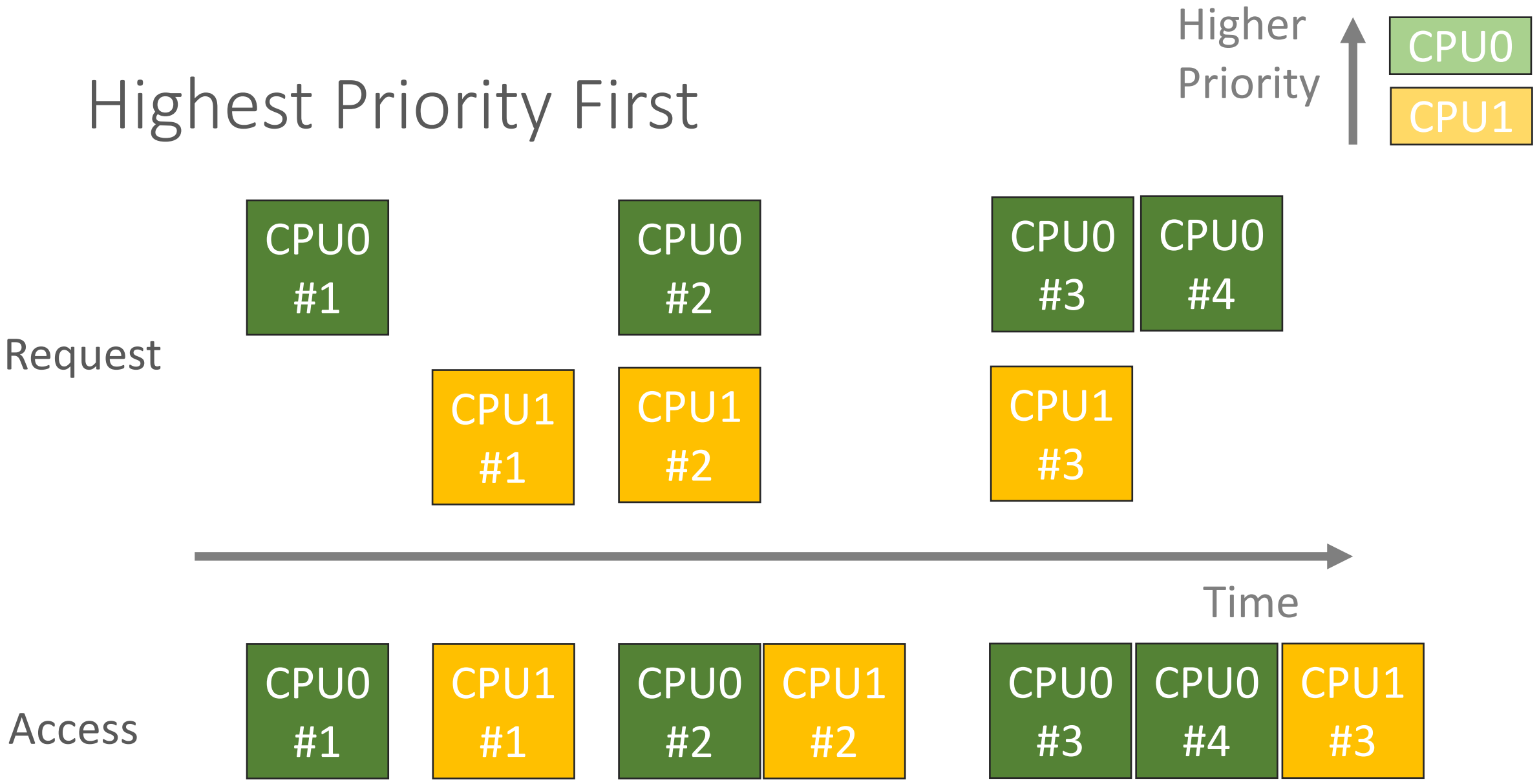
Highest Priority First



Highest Priority First

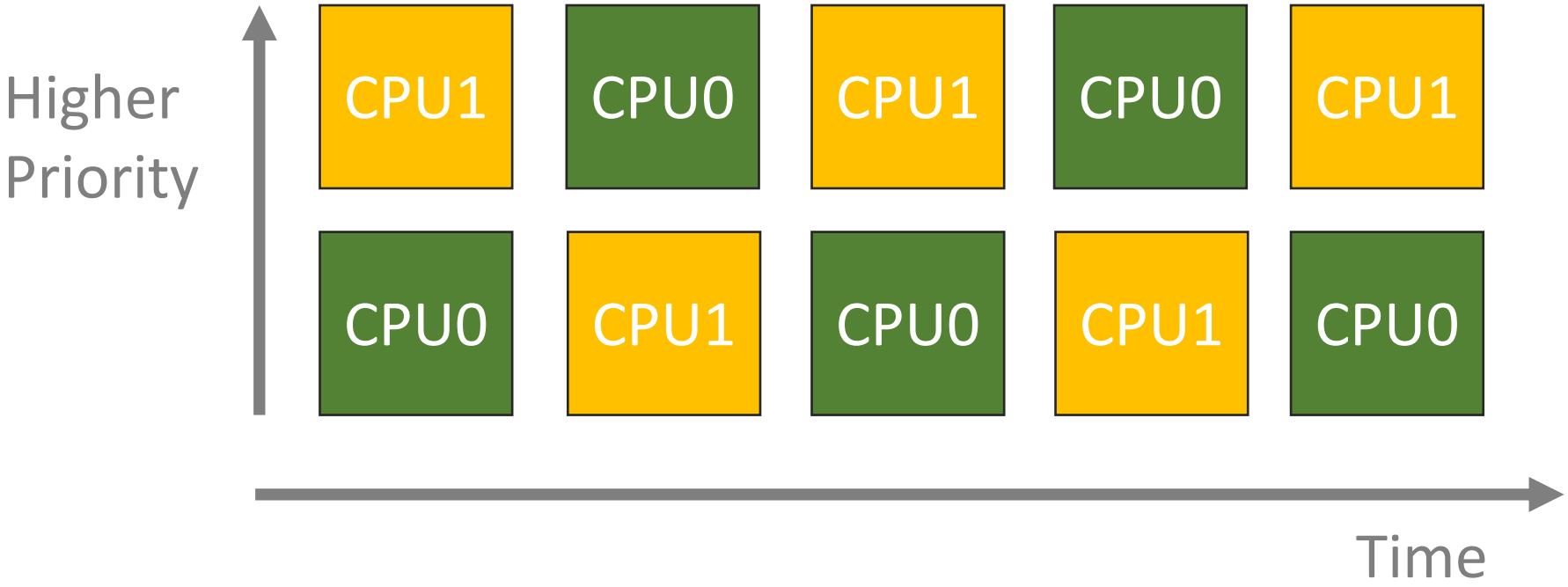


Highest Priority First

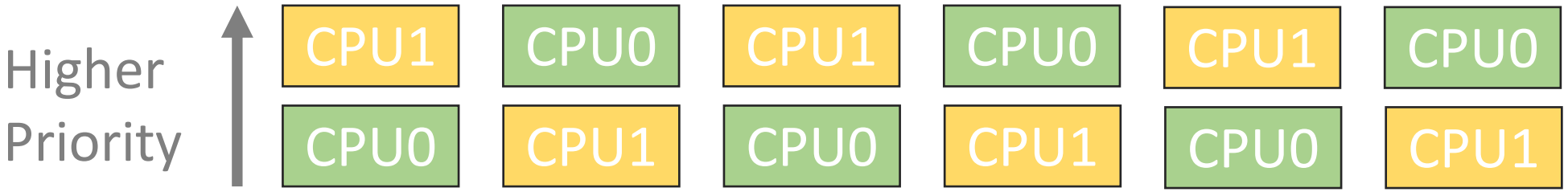


Round Robin

- Priority updates every cycle. Everyone get's equal access to highest priority



Round Robin

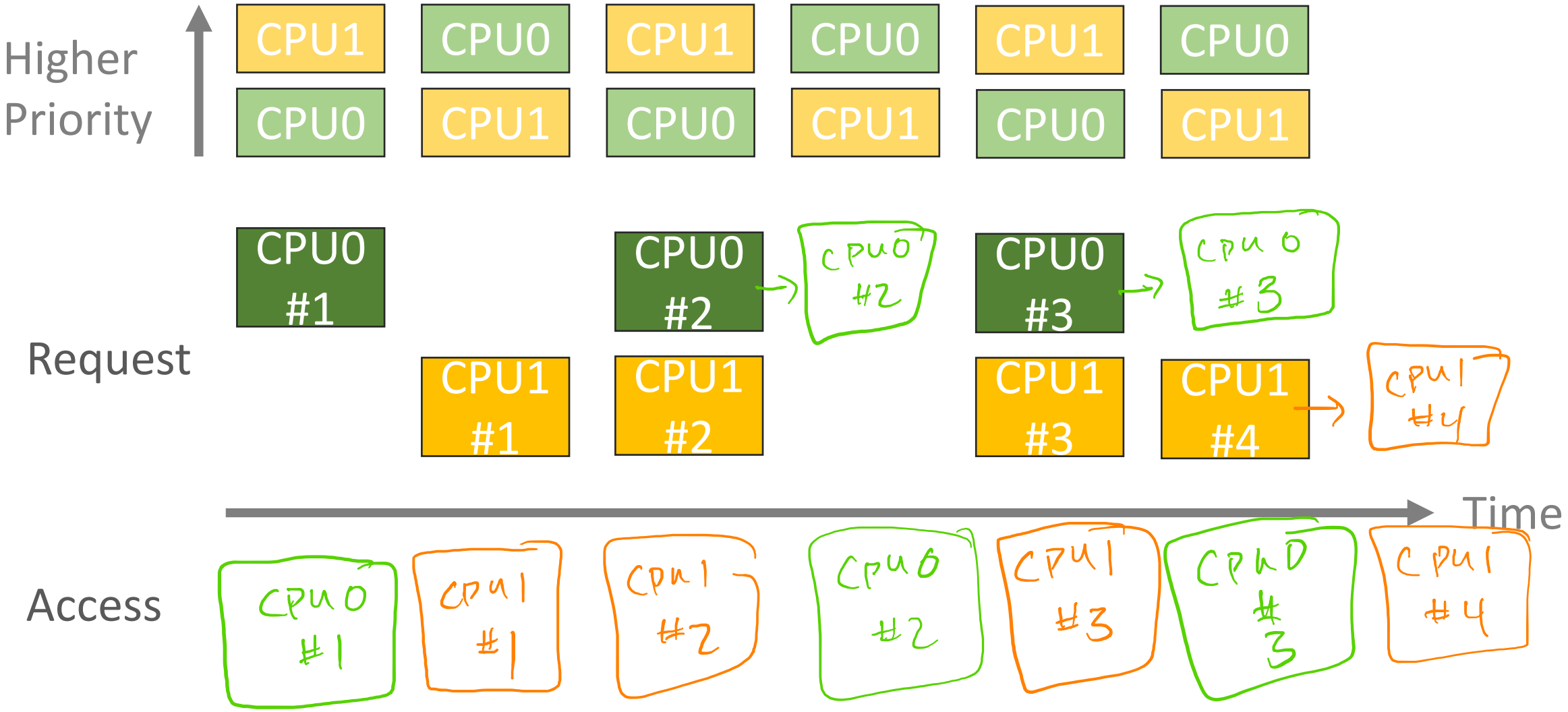


Request

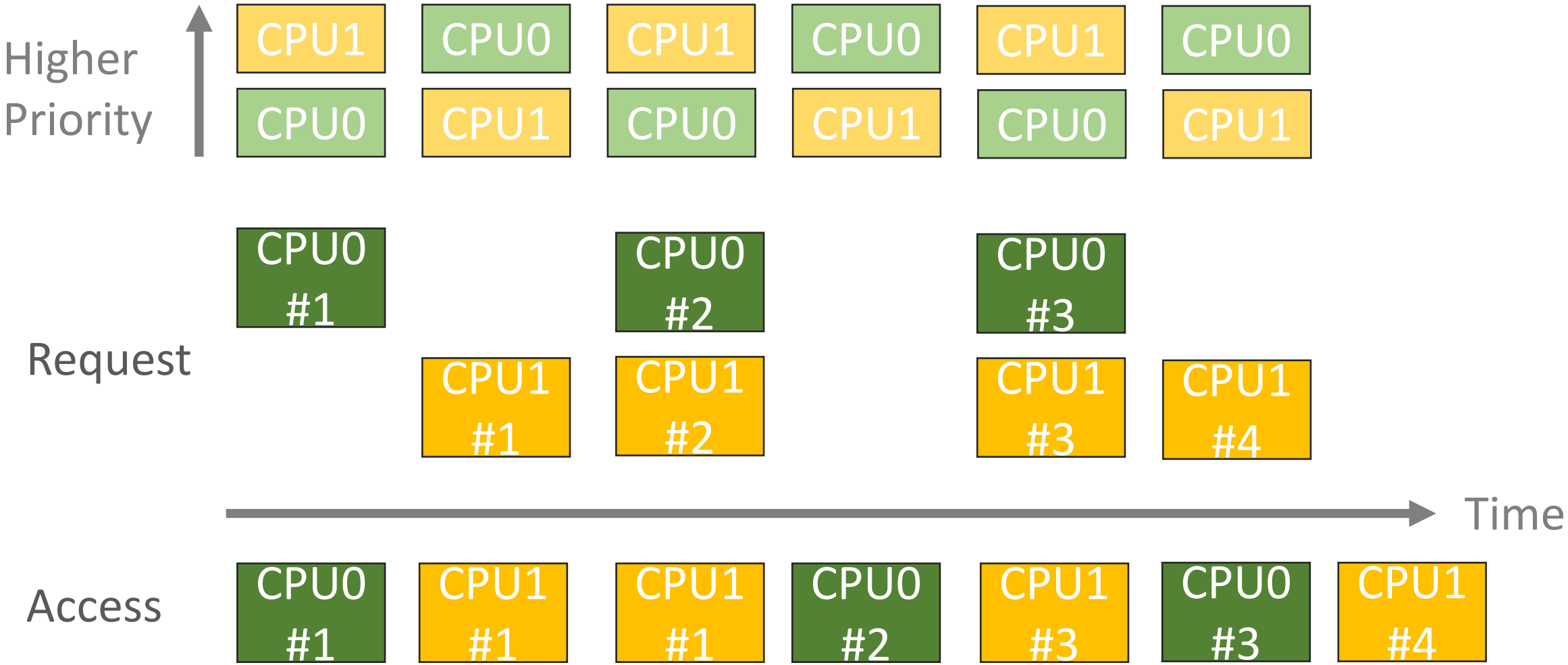
Time

Access

Round Robin


Higher Priority
CPU1
CPU0
CPU1
CPU0
CPU1
CPU0
CPU0
CPU1
CPU0
CPU1
CPU0
CPU1
Request
CPU0 #1
CPU0 #2
CPU0 #2
CPU0 #3
CPU0 #3
CPU1 #1
CPU1 #2
CPU1 #3
CPU1 #4
CPU1 #4
Time
Access
CPU0 #1
CPU1 #1
CPU1 #2
CPU0 #2
CPU1 #3
CPU0 #3
CPU1 #4

Round Robin


Higher Priority
CPU1
CPU0
CPU1
CPU0
CPU1
CPU0
CPU0
CPU1
CPU0
CPU1
CPU0
CPU1
Request
CPU0 #1
CPU0 #2
CPU0 #3
CPU1 #1
CPU1 #2
CPU1 #3
CPU1 #4
Time
Access
CPU0 #1
CPU1 #1
CPU1 #1
CPU0 #2
CPU1 #3
CPU0 #3
CPU1 #4

Other Arbitration ideas?

→ random

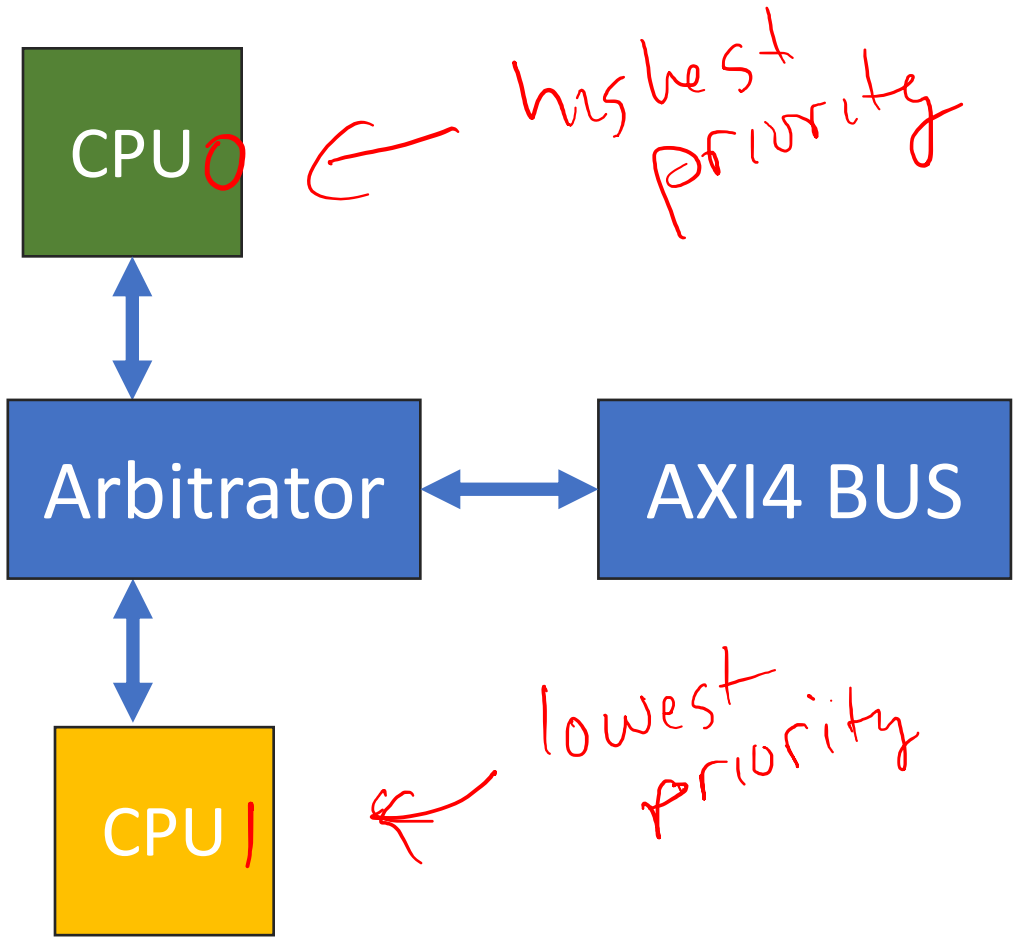
→ spilt priority → 1 high-priority
rest are RR

→ tiered priority

time - weighted

time-weighted random

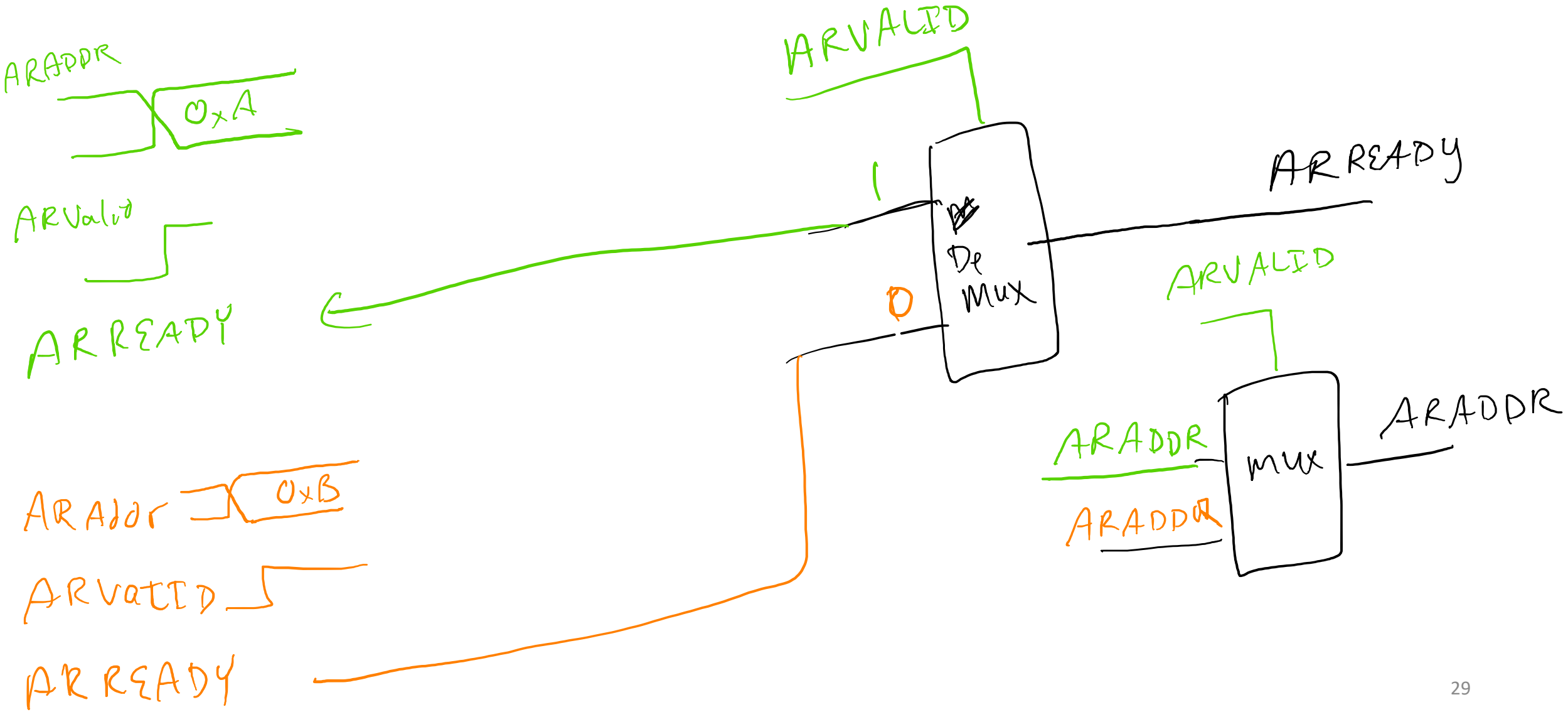
An Arbitrator selects who gets to use the bus



- What happens if both request a transaction at the same time?

- Arbitration:
  - Fixed-Priority
  - Round Robin
  - Many more…

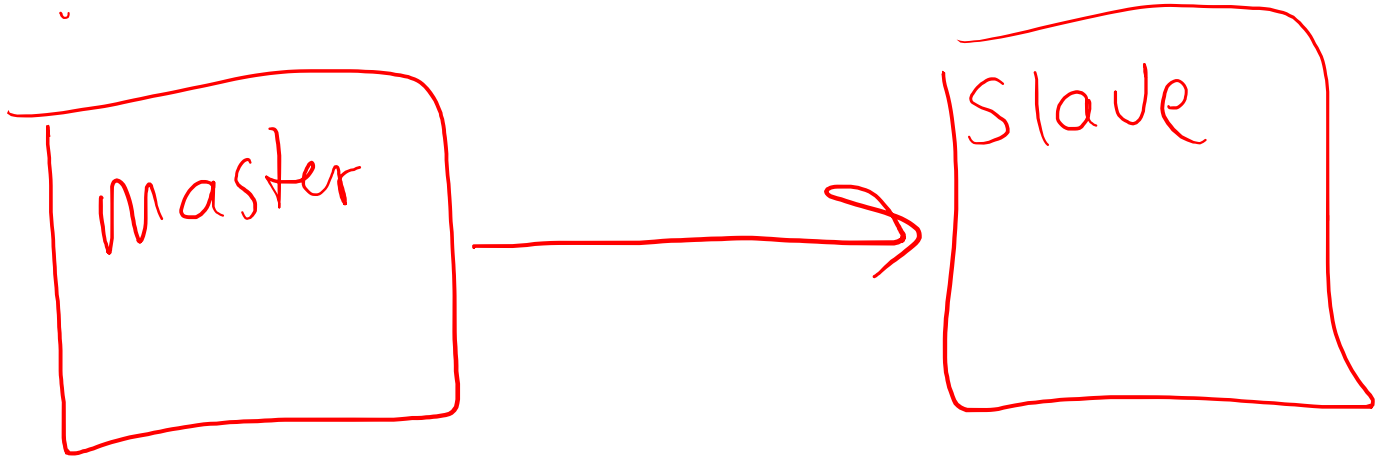
Arbitrator Circuit?

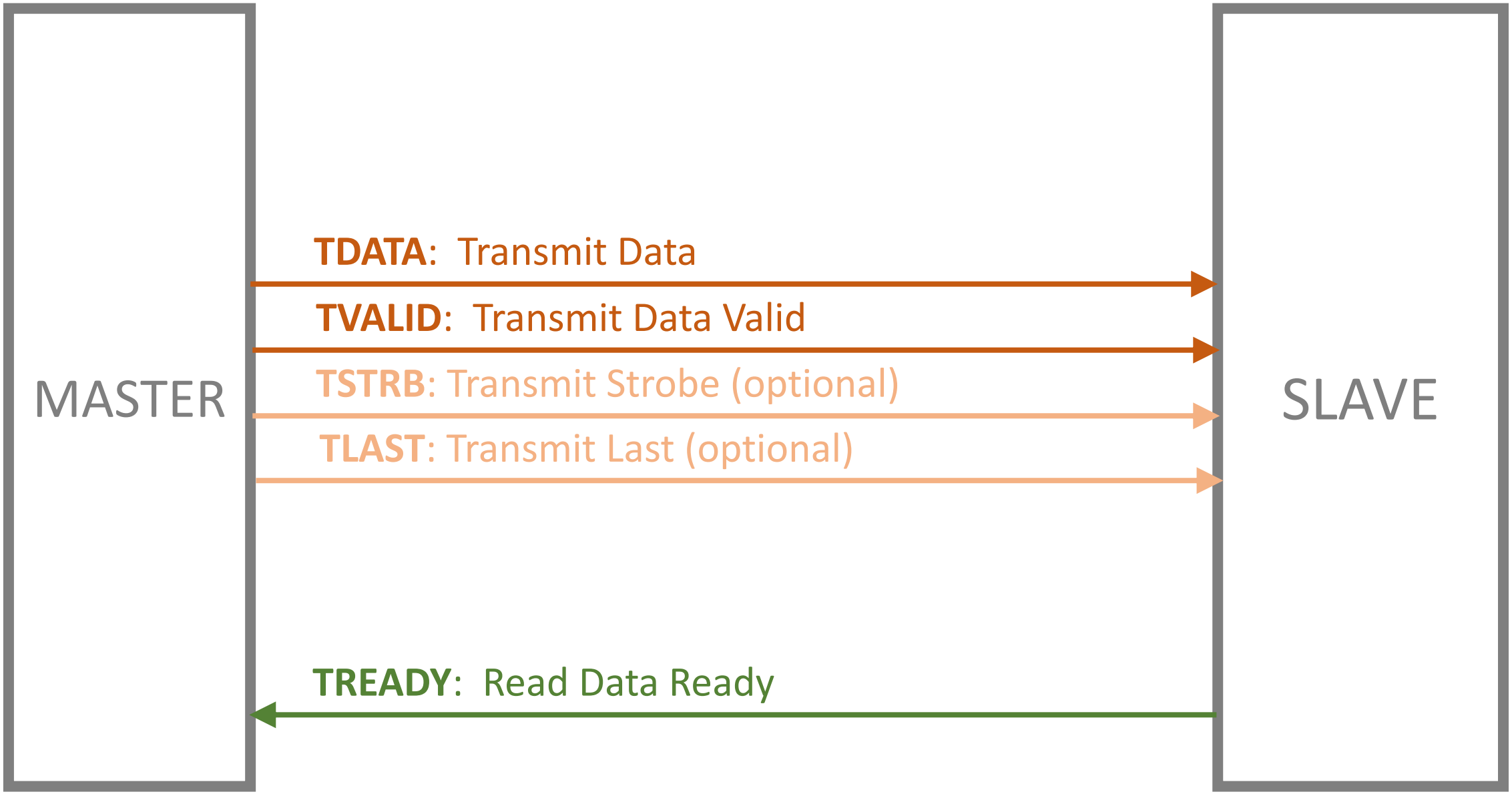


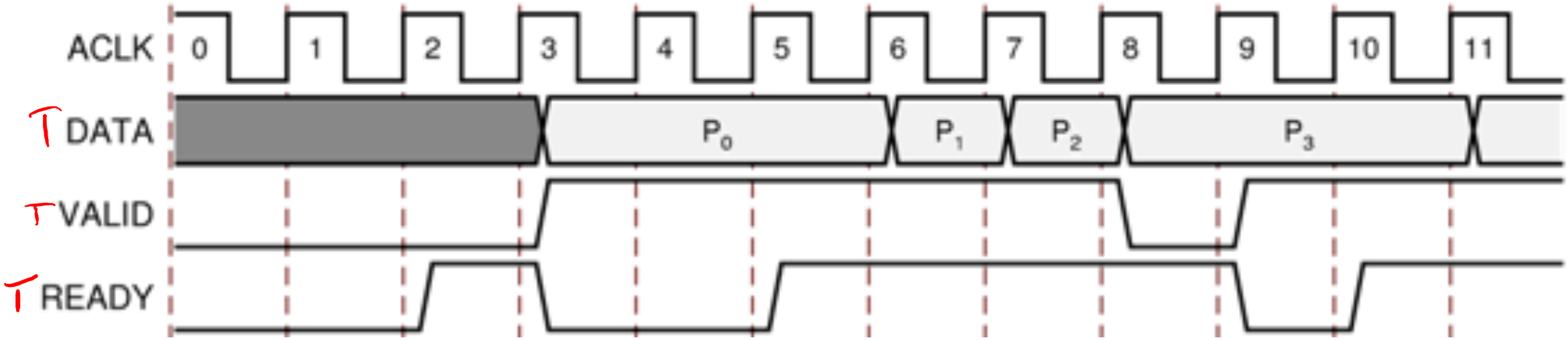
AXI4 Stream

AXI4-Streams

- Simple point-to-point high-speed bus
- NOT MEMORY-MAPPED!




MASTER
TDATA: Transmit Data
TVALID: Transmit Data Valid
TSTRB: Transmit Strobe (optional)
TLAST: Transmit Last (optional)
TREADY: Read Data Ready
SLAVE


ACLK
0
1
2
3
4
5
6
7
8
9
10
11
DATA
P0
P1
P2
P3
VALID
READY

Next Time:

- Direct Memory Access

References

- Zynq Book, Chapter 19 “AXI Interfacing”
- Practical Introduction to Hardware/Software Codesign
  - Chapter 10
- AMBA AXI Protocol v1.0
  - http://mazsola.iit.uni-miskolc.hu/~drdani/docs_arm/AMBAaxi.pdf
- https://lauri.võsandi.com/hdl/zynq/axi-stream.html

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